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THEORETICAL VARIANTS OF EVOLUTION OF ECONOMY AND POPULATION CHANGES

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Abstract. *The article deals with the issues of classification different kinds of countries using surfaces which their economy and population can occupy. Some formulas which are presented in this article allow us to calculate these surfaces.*

Economic growth or recession are affected by a large number of factors one of which is growth or decrease in population for objective reasons. In order to find out the interconnection between theoretical economic growth or recession e_t and theoretical growth or decrease in population p_t respectively, we developed Table 1, where possible variants of interconnection of variables e_t and p_t in question are presented. In this work and for our case the word *economy* will be understood as GDP in the country in question. Thus, depending on context we will use the term *GDP* or the word *economy*. The graphs in Table 1 have one horizontal axis Q_t instead of two Q_{et} and Q_{pt} with different scales and values. This Table contains 62 lines some of them are presented in Table 1.

Hereinafter in all the graphs of Table 1, a dashed line indicates the e_t parameter, and the full line indicates the p_t parameter.

Variant 1

In this variant economy e_t and population p_t are constant values and do not change in time, i.e. $e_t = const$ and $p_t = const$ (see line 1). Hereinafter the first digit indicates the line number, and the second the column number. Changes of e_t and p_t are described by a linear equation of the form $y_{et} = Q_{et}$ and $y_{pt} = Q_{pt}$.

Box 11 presents a variant where the population Q_{pt} prevails over the economic variables Q_{et}, i.e. $Q_{pt} > Q_{et}$. In order to characterize this state of economy and population let us introduce the variable Q_{pet}, which shows theoretical difference between variables Q_{pt} and Q_{et}, and is calculated according to the following formula (1)

$$Q_{pet} = Q_{pt} - Q_{et}. \tag{1}$$

In this case the bigger the Q_{pet} value, the lower the population's living standard, as consumption per capita decreases. For the sake of simplicity let us consider that both the elite and oligarchs and the middle class and low-income people consume the same number of goods, products and services. If we do not take it into account, along with the growth of the Q_{pet} value the elite class and the oligarchs will at least not want to worsen their wealth, so the living standard of the middle class and the low-income will decrease.

Thus, here we may have the following variants of situation change or their combinations:

1. The population will take worsening of its situation lying down;
2. Due to its inescapable situation the population will join different religious sects, which will make its economic situation even worse;
3. The population will emigrate legally and illegally whenever possible to more economically developed countries, primarily to countries where population birth rate is going down and that do not have enough workforce, which can be currently seen in Europe;
4. The government will organize more of various holidays and carnivals to distract the population from real life;
5. The government of the country will pursue the "enemy from without or within" policy to reorient the population's views from the economic situation to politics pointing them out as the cause of the current situation. This can be seen in recent and new history of countries, including in the USSR;
6. The government of the country will pursue the policy of making the population drunk, as this is much easier, quicker and cheaper to do than build stadiums, sports grounds and promote a healthy lifestyle, as drunk population is easier to manipulate and it will not take active part in political life of the country.

Moreover, for example, in the Soviet Union income from vodka sales brought much profit to the public purse. Thus, the population will degrade, which can lead to irreversible effects on the nation's gene pool;

Table 1. Variants of interconnection of theoretical economy changes e_t and theoretical population changes p_t

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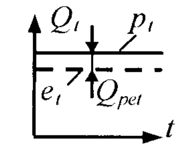
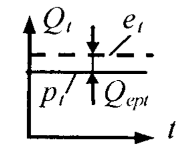
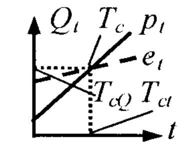
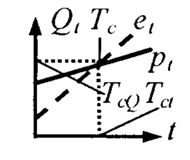
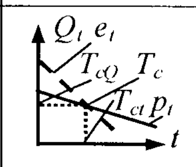
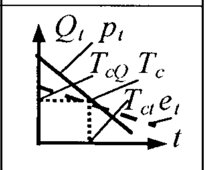
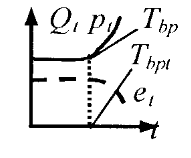
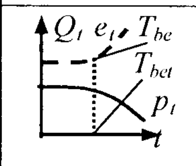
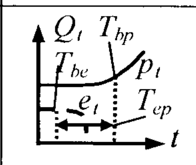
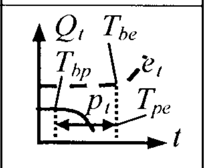
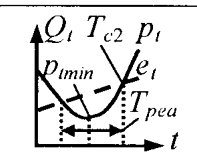
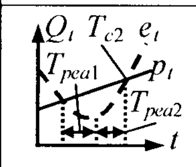
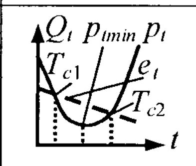
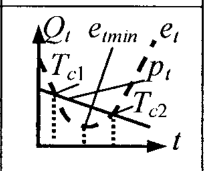
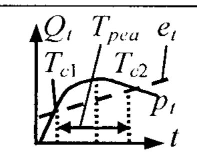
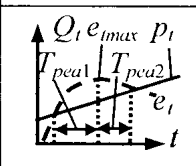
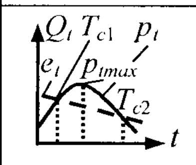
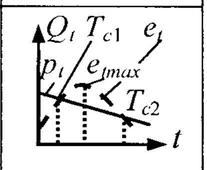
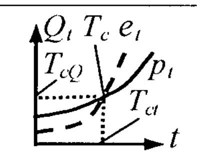
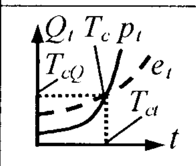
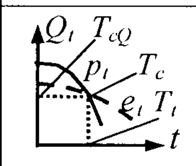
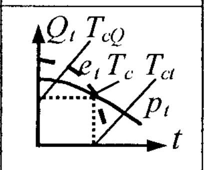
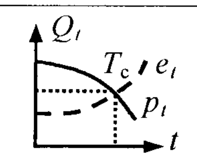
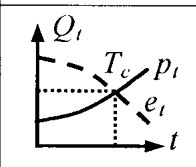
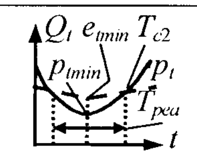
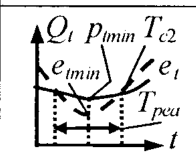
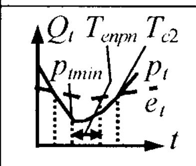
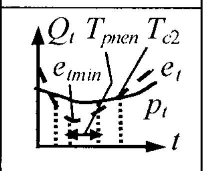
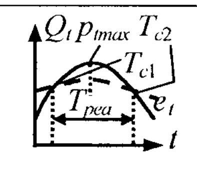
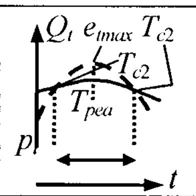
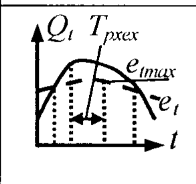
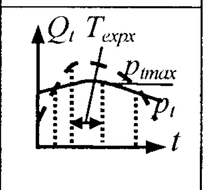
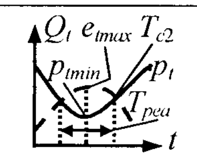
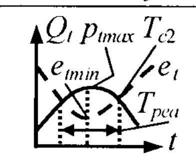
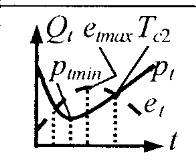
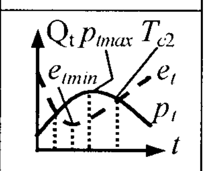
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

7. The government of the country will pursue the policy of "stupefying" the young promoting youth subculture to distract them from active political life;
8. The government of the country will get rid of opposition leaders and activists by different means: by judiciary means, sending them to prison, camps, etc., by physical liquidation, forced emigration to other countries, etc. In the latter case opposition leaders will create foreign centers and, using mass media and the Internet, will start to undermine the antinational regime their compatriots still live in;
9. The government will expel active population from cities to the country on a massive scale as reeducation by physical labor;
10. The government of the country will put the economy on a militaristic footing. This will ultimately lead to worsening of the population's living and possibly to war, which happened in Germany in the last century;
11. In order to distract the population from the democratic process the government of the country will start a war with a weaker neighboring state, which was done by Iraq in 1990 to 1991 and by other countries;
12. In order to improve its own economy the government of the country will join military action started by another state, which can be seen in the history of America, where large GDP growth is directly related to arms and product supply to countries at war;
13. The population will remove the current government and possibly the ruling class from power by means of a democratic election, as it was done in Chili, Venezuela, etc., or by means of a "color revolution" (Ukraine, Georgia);
14. The population will start a revolution and remove the existing regime, as it recently took place in Romania, Tunisia and in other countries.

Box 12 shows dependences where economic variables prevail over population, i.e. $Q_{et} > Q_{pt}$. In this case the higher the Q_{ept} value calculated according to the formula (2), the higher the living standard of the population, which will naturally lead to wealth growth of the whole population.

$$Q_{ept} = Q_{et} - Q_{pt}. \quad (2)$$

Dependences e_t and p_t changing according to the linear law can be characterized by the tilt angle α of a certain line to the horizontal axis. In the variant in question for all cases of $\alpha = 0°$. Theoretically the tilt angle α_t may lie within the following limits $-90° \leq \alpha_t \leq +90°$. Thus, the symbol of the angle α shows growth (if +) or decrease (if –) of the variables in question.

Variant 2

Line 2 shows variants of change of the economy e_t and population p_t according to the linear law, in this case the set up dependences intersect in point T_c, which has values T_{ct} and T_{cQ} at the corresponding reference axes t and Q_t.

In the case in question there are the following four evolution variants possible:

- population p_t grows faster than the economy e_t, box 21. In this case it is desirable that value T_{ct} be at its maximum, which will allow to postpone impoverishment of the country's population;
- the economy e_t grows faster than population p_t, box 22. In this case it is desirable that value T_{ct} be at its minimum. This will allow to increase population's wealth in a shorter time period;
- the economy et decreases faster than population pt, box 23. In this case it is desirable that value Tct be at its maximum. This will prevent population from fast impoverishment and allow to take emergency measures to both increase the country's population and improve the economy as a whole;
- population p_t decreases faster than the economy e_t, box 24. In this case it is desirable that value T_{ct} be as low as possible. This will allow to increase population's wealth in a minimum time period. This variant is not entirely ethical from the moral viewpoint, as sensible population will not want to increase its wealth at the expense of another part of the country's population, moreover, as population decreases, the workforce producing common goods and services decreases accordingly. That is why in this variant it is desirable to use automation and mechanical upgrade of production process, as well as introduction of new technologies, etc. This variant is relevant for countries with large population, poor economic situation and thus excess of workforce.

Variant 3

Here variant 3, which was inserted into this group, though it could have been included into the group of nonintersecting curves. In this variant variables e_t and p_t are parallel to axis t at first, and then they start changing according to a complex law in the points of bend of T_{be} and T_{bp}, where the tangent changes its direction, with similar coordinates (boxes 31 and 32), or with different coordinates (boxes 33 and 34). These boxes also show projections of points T_{be} and T_{bp} to axis t, which are T_{bet} and T_{bpt} respectively. Due to the fact that the coordinates of points T_{be} and T_{bp} are different, we introduced here two new properties of T_{ep} and T_{pe}, which are calculated with the formulas: $|T_{ep}| = T_{bet} - T_{bpt}$; $|T_{pe}| = T_{bpt} - T_{bet}$. Here the values of T_{ep} and T_{pe} are considered according to the absolute value, as they can be the following: $T_{bpt} > T_{bet}$; $T_{bpt} < T_{bet}$ or $T_{bpt} = T_{bet}$.

On the basis of the formulas above we can conclude the following:

- if the economy e_t and population p_t change their linear law to a more complex one at the same time, i.e. $T_{bet} = T_{bpt}$, as it is presented in boxes 31 and 32, let us add new parameters in order to make conclusions: $\pm\Delta e_t$; $\pm\Delta p_t$; Δt_e and Δt_p, where $\pm\Delta e_t$ is growth or recession of the economy in the time period Δt_e in question, unit, and $\pm\Delta p_t$ is growth or decrease of population in the time period Δt_p in question, pers. This means that we need to know how variables e_t and p_t will increase or decrease within the time period Δt_e or Δt_p. In this case values Δt_e and Δt_p should be taken as equal, i.e. $\Delta t_e = \Delta t_p$. In effect it is desirable to assume that values Δt_e and Δt_p are equal to one, i.e. one hour, day, week, month or year, depending on specific conditions and the prospect of timely receiving accurate information on changes taking place in the economy or population.

Thus, we can note down the following three formulas for box 31:

1. $\Delta p_t - |-\Delta e_t| = 0$;
2. $\Delta pt - |-\Delta et| > 0$;
3. $\Delta p_t - |-\Delta e_t| < 0$.
4.

As for box 32, we can note down the following:

5. $\Delta et - |-\Delta pt| = 0$;
6. $\Delta et - |-\Delta pt| > 0$;
7. $\Delta e_t - |-\Delta p_t| < 0$.
8.

For the sake of simplicity here the + sign was omitted.

In boxes 31 and 32 in question there are the following two evolution variants possible:

1. When variables e_t and p_t change by the same value, i.e. when $|\pm\Delta e_t| = |\pm\Delta p_t|$. If the values are equal $|\pm\Delta e_t| = |\pm\Delta p_t|$, it is desirable for box 91 that the value of point T_{bpt} be at its maximum, as in this case the population's wealth will decrease far in the future. As for values $|\pm\Delta e_t|$ and $|\pm\Delta p_t|$, here they must be at their minimum, as in this case the population's wealth will decrease very slowly;

2. When variables e_t and p_t change according to different laws, i.e. $|\pm \Delta e_t| \neq |\pm \Delta p_t|$. Here it is desirable for variables e_t and p_t in box 31 that the sum of values $|-\Delta e_t|$ and Δp_t be at its minimum $(|-\Delta e_p| + \Delta p_t) \rightarrow min$. In this case population growth and simultaneous recession of the country's economy will take place more smoothly. As for box 32, here it is desirable to have maximum Δe_t and $|-\Delta p_t|$ values, as in this case population's wealth will increase in a minimum time period.

For boxes 33 and 34 there are the following two evolution variants possible:

1. If the economy starts falling earlier than population starts growing, it is desirable that the value of T_{ep} tend to the maximum, i.e. $T_{ep} \rightarrow max$, as in this case population will slide into poverty slower (box 33);

2. If the economy starts improving later than population starts decreasing, it is desirable that the value of T_{pe} tend to the minimum, i.e. $T_{ep} \rightarrow min$, as in this case growth of population's income may have a good impact on birth rate (box 34).

Variant 4

Line 14 shows variables e_t and p_t that differ from those above in that one of the dependences is a curve that goes down to its minimum T_{emin} (T_{pmin}) at first, and then goes up intersecting the line twice. In boxes 41 and 42 the lines in question go up, and in boxes 43 and 44 they go down.

Here there must be the following values of the parameters in question.

For parameters Δe_t and Δp_t:

- boxes 41, 43:

to the minimum point: $-\Delta p_t \rightarrow max$;

beyond the minimum point: $\Delta p_t \rightarrow min$,

- boxes 42, 44:

to the minimum point: $-\Delta e_t \to min$;

beyond the minimum point: $\Delta e_t \to max$.

For parameters T_{pea}:

- box 41: $T_{pea} \to max$, $T_{pea1} \to max$, $T_{pea2} \to max$;
- box 42: $T_{pea} \to min$, $T_{pea1} \to min$, $T_{pea2} \to min$;
- box 43: $T_{pea} \to max$, $T_{pea1} \to max$, $T_{pea2} \to max$;
- box 44: $T_{pea} \to min$, $T_{pea1} \to min$, $T_{pea2} \to min$.

In this variant while choosing the maximum or minimum values of the parameters in question we encountered duality of their interpretation. So, for example, if in box 44 parameter T_{ct1} is at its maximum, population's economic situation will decrease for a long time, and then it will shoot up. If the value of parameter T_{ct1} is at its minimum, population's economic situation will shrink, but it will shoot up then. In this case the author, of course, considers it desirable to choose the second evolution variant, where $T_{ept1} \to min$.

Variant 5

This variant is similar to variant 4, but here the curve has a maximum. That said, at first the curve goes up to its maximum T_{emax} (T_{pmax}), and then goes down and intersects the line twice. In boxes 51 and 52 the lines in question go up, and in boxes 53 and 54 they go down.

Here there must be the following values of the parameters in question:

For parameters Δe_t and Δp_t:

- boxes 51, 53:

before the maximum point: $\Delta p_t \to min$;

beyond the maximum point: $-\Delta p_t \to max$

- boxes 52, 54:

before the maximum point: $\Delta e_t \to max$;

beyond the maximum point: $-\Delta e_t \to min$.

For parameters T_{pea}:

- box 51: $T_{pea} \to min$, $T_{pea1} \to min$, $T_{pea2} \to min$;
- box 52: $T_{pea} \to max$, $T_{pea1} \to max$, $T_{pea2} \to max$;
- box 53: $T_{pea} \to min$, $T_{pea1} \to min$, $T_{pea2} \to min$;
- box 54: $T_{pea} \to max$, $T_{pea1} \to max$, $T_{pea2} \to max$.

Variants 6 and 7

These two variants show variables e_t and p_t that intersect in one point T_c.

Here there must be the following values of the parameters in question:

- box 61: $T_{ct} \to min$, $\Delta e_t \to max$, $\Delta p_t \to min$;
- box 62: $T_{ct} \to max$, $\Delta e_t \to max$, $\Delta p_t \to min$;
- box 63: $T_{ct} \to min$, $-\Delta e_t \to min$, $-\Delta p_t \to max$;
- box 64: $T_{ct} \to max$, $-\Delta e_t \to min$, $-\Delta p_t \to max$;
- box 71: $T_{ct} \to min$, $\Delta e_t \to max$, $-\Delta p_t \to max$;
- box 72: $T_{ct} \to max$, $-\Delta e_t \to min$, $\Delta p_t \to min$.

The following three variants 8, 9 and 10 show intersections of curves in two points, and in this case the maximum and minimum coordinates of variables e_t and p_t can be equal, i.e. $T_{ptmin} = T_{etmin}$ etc. (boxes), or different, i.e. $T_{ptmin} > T_{etmin}$ or $T_{ptmin} < T_{etmi}$, etc.

Variant 8

In order to describe variables e_t and p_t shown in line 8, as well as in lines 9 and 10, which can have their maximum and minimum values both at one line and dislocated, let us introduce the following eight new parameters given below:

1. T_{pnen} is the difference between the coordinates of points T_{ptmin} and T_{etmin}. Unit;
2. T_{enpn} is the difference between the coordinates of points T_{ptmax} and T_{etmax}. Unit;
3. T_{pxex} is the difference between the coordinates of points T_{ptmax} and T_{etmax}. Unit;
4. T_{expx} is the difference between the coordinates of points T_{etmax} and T_{ptmax}. Unit;
5. T_{pxen} is the difference between the coordinates of points T_{ptmax} and T_{etmin}. Unit;
6. T_{expn} is the difference between the coordinates of points T_{etmax} and T_{ptmin}. Unit;
7. T_{pnex} is the difference between the coordinates of points T_{ptmin} and T_{etmax}. Unit;
8. T_{enpx} is the difference between the coordinates of points T_{etmin} and T_{ptmax}. Unit;

The eight parameters described above show dislocation of the peaks of variables e_t and p_t. This said, the subscript *enpx* shows that the first intersection point is a curve, in this case a coordinate of the minimum point of variable e_t, and the second intersection point is the maximum point of variable p_t, i.e. $T_{etmin} > T_{ptmax}$.

Line 21 shows variants of change of variables e_t and p_t in the form of two curves having minimum points and intersecting in two points. Here in boxes 81 and 82 their minimums are located at one line, and in boxes 83 and 84 they are dislocated in relation to each other by the value of T_{enpn} (box 83) or by T_{pnen} (box 84).

Here there must be the following values of the parameters in question.

For parameter T_{pea}:

- boxes 81 and 83: $T_{pea} \to max$;
- boxes 82 and 84: $T_{pea} \to max$.

For parameters T_{enpn} and T_{pnen}:

- boxes 81 and 83: $T_{pnen} \to min$ where $T_{bet} > T_{bpt}$;

$T_{enpn} \to max$ where $T_{bet} < T_{bpt}$

For parameters Q_{enpn} and Q_{pnen}:

- boxes 81 and 83: $Q_{enpn} \to max$;
- boxes 82 and 84: $Q_{pnen} \to min$;

Variant 9

Line 9 shows variants of change of variables e_t and p_t as two curves having maximum points and intersecting in two points. Here in boxes 91 and 92 their maximums are located at one line, and in boxes 93 and 94 they are dislocated in relation to each other by the value of T_{pxex} (box 93) or by T_{expx} (box 94).

Here there must be the following values of the parameters in question.

For parameters T_{epx} and T_{pex}:

- boxes 91 and 93: $T_{pex} \to min$;
- boxes 92 and 94: $T_{epx} \to max$.

For parameters T_{expx} and T_{pxex}:

- box 93: $T_{pxex} \to max$ where $T_{bet} > T_{bpt}$;

$T_{expx} \to min$ where $T_{bet} < T_{bpt}$;

- box 94: $T_{pxex} \to min$ where $T_{bet} > T_{bpt}$;

$T_{expx} \to max$ where $T_{bet} < T_{bpt}$.

For parameters Q_{expx} and Q_{pxex}:

- boxes 91 and 93: $Q_{pxex} \to min$.
- boxes 92 and 94: $Q_{expx} \to max$.

Variant 10

Line 10 shows variants of change of variables e_t and p_t, as two curves having maximum points and intersecting in two points. Here in boxes 101 and 102 their maximums and minimums are located at one line, and in boxes 103 and 104 they are dislocated in relation to each other by the value of T_{pxex} (box 103) or by T_{expx} (box 104).

Here there must be the following values of the parameters in question.

For parameters T_{epx} and T_{pex}:

- boxes 101 and 103: $T_{pex} \to max$
- boxes 102 and 104: $T_{epx} \to min$

For parameters T_{enpx} и T_{pnex}:

- box 103: $T_{pnex} \rightarrow max$ where $T_{bet} > T_{bpt}$;

$T_{pnex} \rightarrow max$ where $T_{bet} < T_{bpt}$;

- box 104: $T_{enpx} \rightarrow min$ where $T_{bet} > T_{bpt}$;

$T_{enpx} \rightarrow min$ where $T_{bet} < T_{bpt}$,

For parameters Q_{pxen} и Q_{expn}:

- boxes 101 and 103: $Q_{expn} \rightarrow max$;
- boxes 102 and 104: $Q_{pxen} \rightarrow min$.

Due to the fact that two variables e_t and p_t have the same coordinates when they are tangent, let us call it *equilibrium line* L_{bal}. This line has such properties as the following: population consumes entirely all products, goods and services produced in the country in the time period of them being tangent.

Based on the tangent and merging variables e_t and p_t described above, let us introduce concepts in a similar way to the *equilibrium point*: *equilibrium of the economy and population*, i.e. where the GDP value and population are equal $Q_{eti} = Q_{pti}$.

Now let us introduce the following three definitions to *equilibrium of the economy and population.*

Theoretical equilibrium of the economy and population is understood to be the equation $Q_{eti} = Q_{pti}$, *where the population consumes all the quantity of products, goods and services in a time unit its country produces in the same time unit without loss, and import is fully absent.*

Practical equilibrium of the economy and population is understood to be the equation $Q_{epi} = Q_{ppi}$, *where the population consumes all the quantity of products, goods and services in a time unit its country produces in the same time unit considering their loss* ΔQ_{etl}, *and import is fully absent.*

Real equilibrium of the economy and population is understood to be the inequation $Q_{eri} = Q_{pri}$, *where the population consumes all the quantity of products, goods and services in a time unit its country produces in the same time unit considering their loss* ΔQ_{etl}, *and the remaining lack of products, goods and services in covered by import.*

SPECIALTY OF THE USAGE OF ELECTRONIC MARKETING IN TOURISM

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Abstract. *In the given work there were studied benefits of using electronic marketing in sphere of tourism. Contrary to the traditional communication media which basic function consists in information delivery, the Internet is not simply the information transmitter, and much more - the global virtual market. Tourism today is the global computerized business in which participate air companies, hotel chains and tourist corporations of the entire world. Modern tour product becomes more flexible and individual, more attractive and accessible to consumer. Electronic marketing is accomplishing electronic function by the Internet, information technologies and telecommunication systems. Electronic marketing is also called internet marketing, web marketing, online marketing or digital marketing.*

Keywords: *Internet, electronic marketing, display advertising, eE-mail marketing, blog marketing, social media marketing, mobile marketing.*

Today`s society is in such stage that the information technologies are being rapidly spread and the influence of it is enough considerable in most of part of economic life.

Availability of electronic atmosphere by the result of the development of global information and communicative technologies and the large number of companies using the opportunities of this atmosphere in their working process is the most significant mark of economics. Todays` businessmen are actively using opportunities of information technologies. In most cases, the usage degree of it signs the success of running business, and its place in inner and outer world market.